MILITARY
SPECIFICATION

MIL-T-55046A(SigC)
14 January 1960
Superseding
MIL-T-55046(SigC)
25 November 1958

TRANSPONDER SET AN/APX-44( )

1. SCOPE

1.1 This specification covers one type of equipment designated as Transponder Set AN/APX-44( ), and the following units thereof:

Units

a. Receiver-Transmitter RT-494( )/APX-44

b. Control, Transponder Set C-2714( )/APX-44

c. Antenna AT-884( )/APX-44

d. Mounting MT-2100( )/APX-44

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

TT-P-141	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for.
MIL-T-5422	Environmental, Testing, Aircraft Electronic Equipment.

FSC 5895

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MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys.
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment.
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting.
MIL-E-7894	Electrical Power, Aircraft Characteristics of.
MIL-M-13231	Marking of Electronic Items.
MIL-P-13949	Plastic Sheet, Laminated, Copper-Clad Paper Base Phenolic.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-252	Wire Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-429	Printed-Circuit, Terms and Definitions.
MIL-STD-595	Colors.

DRAWINGS

SIGNAL CORPS

SC-A-362100	Preproduction Pack, Preparation of.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.- The set covered by this specification provides transponder functions necessary to receive, decode, and respond to the characteristic IFF interrogations of either the Mark X IFF System or the Mark X System supplemented by the Selective Identification Feature (SIF) (IFF Mark X (SIF). The set is constructed so as to permit installation in smaller type aircraft such as, observation aircraft, helicopters, etc.

3.2 Procurement model.- A procurement model of the equipment on contract will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, the equipment on contract shall conform to the following:

3.2.1 Construction.- Physical construction of the equipment on contract shall conform to the procurement model.

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3.2.2 Features.- The equipment on contract shall incorporate all features of the procurement model.

3.2.3 Performance.- In case of conflict between specified performance characteristics for the items on contract and performance of the model, the former shall govern.

3.3 Transponder Set AN/APX-44( ), General Requirements.- Except as otherwise specified herein, the transponder set shall conform to Specification MIL-E-5400.

3.4 Materials, parts and processes.- The following detail requirements shall apply, in addition to the general requirement of MIL-E-5400.

3.4.1 Finish, protective.- Units, except for the Antenna specified below, shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on type I surfaces shall be black color lusterless enamel matching a color chip provided by the procuring agency. The finish for the Antenna shall be irridite film in accordance with MIL-C-5541, followed by (olive drab) lusterless enamel, color chip number 34087, in accordance with Federal Standard No. 595.

3.4.2 Isolators, vibration.- Vibration isolators of the types listed below, shall be used on Mounting MT-2100( )/APX-44. Holes suitable for mounting Lord type HTO-7 isolators, or equal, shall be also provided in the equipment.

For dimpled trays	For flat trays
Lord Mount J-5572-3 Short stud or Barrymount M-44-AA-10, or equal	Lord Mount J-5572-3 Long stud or Barrymount M-44-BA-10, or equal

3.4.3 Tropicalization.-

3.4.3.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

a. Each component part and subassembly meets one of the following:

(1) It is furnished under a individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in Specification MIL-E-5400 unless the material has been given the treatment specified in Specification MIL-E-5400 or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

b. The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

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3.4.3.2 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This statement shall include the statements required by Specification MIL-T-152.

3.4.4 Marking.-

3.4.4.1 General.- Marking shall conform to Specification MIL-M-13231, except that any group I, marking and any material and process for labels, as described in that specification, may be used for surfaces exposed to view during use or storage of the equipment. Front panel marking shall be group 1.

3.4.4.2 Visibility.- Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.4.3 Serial numbers.- Units as listed in 1.1a through 1.1d inclusive shall be serial numbered.

3.4.5 Materials and processes for printed wiring assembly.-

3.4.5.1 Printed wiring board.-

(a) Etching.- The etching of foil-plastic laminates is the preferred method for fabricating printed wiring boards. Appropriate care and suitable cleaning procedures shall be employed to avoid detrimental contamination of the printed wiring boards.

(b) Base material.- The material used as the base for foil-plastic laminates shall be glass-fabric base, epoxy resin. The metal used shall be copper foil, meeting the requirements of Specification MIL-P-13949. The composite laminate shall meet the minimum quality and performance requirements of Specification MIL-P-13949, as well as any more stringent requirement which may be dictated by a particular application.

(c) Terminal areas.- Terminal areas shall consist of an annular ring conductor pattern completely circumscribing and abutting on the electronic part mounting hole. Wherever possible, minimum overall dimensions of the terminal area shall exceed by at least .095 inch, the corresponding dimensions of the mounting hole; at no point shall the edge of the terminal area be closer than .040 inch to the edge of the mounting hole. The contracting officer's approval will be necessary for smaller terminal area clearances.

(1) Where eyelets are employed on printed wiring boards; the terminal area shall be of sufficient diameter to provide an annular ring extending at least .030 inch beyond the projected flange to provide the requisite area for the formation of a good solder fillet.

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3.4.5.2 Electronic part mounting.-

(a) Lead-mounted parts.- Electronic parts shall be mounted and leads attached in such a manner as to avoid the possibility that the bond between the foil and the plastic base may be stressed in either torsional or tensional shear, either during assembly or during subsequent operation and maintenance.

(b) Axial lead shaping.- Leads of axial-leaded parts shall remain straight for at least .015 inch from the part extremity. An additional straight portion of lead shall be provided to compensate for normal variations in part length plus overall machine handling clearances of not greater than .020 inch. The inner radius of bend of lead shall not be less than .045 inch.

(c) Mounting holes.- Cross sectional dimensions of mounting holes for leads of dip-soldered electronic parts shall not exceed by more than .015 inch the corresponding cross sectional dimensions of the leads. A separate mounting hole shall be used for each lead.

(d) Eyelets.- The use of mechanical eyelets or standoff terminals for electrical connections shall be avoided wherever possible because of difficulties in assuring reliable performance. Where frequent replacement of a particular electronic part is anticipated, suitable mechanical eyelets or plated-through holes may be used to prevent delamination of the printed conductor. Mechanical eyelets for such purposes shall be of light-gauge metal and shall be mechanically secured to the printed wiring board. Their flanges shall be flared out from the printed wiring so as to ensure the formation of an adequate solder fillet under the flange. The use of eyelets may make replacement of the part difficult or impossible unless the part leads can be individually and separately removed from the printed wiring board.

(e) Plated-through holes.- Plated-through holes shall not be used by themselves to effect connection between opposite faces of printed wiring boards. A properly formed lead shall be used for this purpose.

(f) Leads, after insertion.- Leads, after insertion through the printed wiring board, shall be clinched, flared, swaged, or otherwise formed, whenever possible, to assist in mechanically anchoring the electronic part during and after assembly. Where straight-through leads must be used, the leads shall extend beyond the foil surface by at least .030 inch.

(g) Soldered joint reheating.- If any portion of an existing printed wiring solder joint is heated to a temperature above 300°F, all solder in the joint shall be reheated to above, the liquidus temperature and the joint reformed to prevent crystallization or oxidation of the joint.

3.4.5.3 Processing.-

(a) Soldering.- Properly controlled dip soldering (time not in excess of 5 seconds and temperature not above 520°F) or other suitable automatic soldering means shall be used wherever practicable for connecting the part terminations to the conductor pattern. Whenever hand soldering is used, suitable equipment and necessary care shall be used to avoid the application of excessive heat to the pattern.

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3.4.5.3 Processing.- (Cont'd)

(b) Flux.- Flux shall be in accordance with Specification MIL-F-14256. The flux shall be used in minimum amounts with the excess removed after the soldering operation. Care shall be taken in the removal of the flux to avoid subjecting the electronic parts to solvents which they were not designed to withstand.

3.4.5.4 Protective insulating coating.- A thin protective coating shall be applied to printed wiring assemblies to protect the electrical insulating properties of the insulating materials used in the printed wiring assembly and to offer some improvement in the mechanical ruggedness of the assembly. This coating shall be either an air-drying or bake-drying epoxy resin, and shall meet the following requirements:

(a) Physical requirements.-

(1) Thickness.- The thickness of the coating on flat areas of the assembly shall be .012 ±.007 inch. Coating buildup on a between components and leads shall not detract from the neatness of the assembly.

(2) Curing time and temperature.- When cured in accordance with the manufacturer's instructions (not to exceed 2 hours at 166°F, maximum), the material shall cure to full hardness, as defined in Method 406.1 of Specification TT-P-141.

(3) Temperature resistance.- No visible deterioration of the coating material, printed conductors, or base material shall occur as a result of the temperature cycling test of 4.13.3.

(4) Moisture resistance.- No visible deterioration of the coating material, printed conductors, or base material shall occur as the result of the moisture resistance test of 4.13.4.

(5) Appearance.- The cured film shall be free from pinholes, bubbles, entrapped contaminants, or other defects, and shall not mask or obliterate the color coding of electronic parts.

(b) Electrical requirements.-

(1) Insulation resistance.- The average of the surface insulation resistance of all specimens before subjection to moisture conditions shall be 50,000 megohms, minimum; no specimen shall be below 25,000 megohms. The average of the surface insulation resistance during and after subjection to moisture conditions as specified in 4.13.4 shall be 100 megohms, minimum; no specimen shall be below 50 megohms.

(2) Dielectric strength.- No disruptive discharge shall occur during the dielectric test of 4.13.6.1.

3.4.6 Radio interference suppression.- The generation of radio interference by the system and the vulnerability of the system to radio interference shall be controlled within the limits of Specification MIL-I-6181. (See 4.16). An external line filter may be used, if necessary to meet the requirements for conducted and radiated interference. When filters are considered necessary, they shall be provided with the equipment.

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3.4.7 Service conditions.- Except as otherwise specified herein, the equipment shall meet the following service conditions (environmental) of Specification MIL-E-5400 shall apply.

3.4.7.1 Operation.- Continuous use for a period of 1 year, under the worldwide environmental conditions specified in 3.4.7.2 through 3.4.7.8, with a duty cycle of 12 hours on and 12 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials and processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.17) shall be considered compliance with this requirement.

3.4.7.2 Temperature.- Altitude combinations. (See 4.11.1).

a. Temperature

(1) Operating.- Continuous operation over the temperature range of +130°F (+55°C) to -65°F (-55°C). Intermittent or continuous operation for at least 30 minutes over the temperature range of +160°F (+71°C) to -65°F (-55°C).

(2) Nonoperating.- Temperature extremes, as specified for class 1 equipment in Specification MIL-E-5400.

b. Altitude

(1) Operating.- Continuous operation at barometric pressures ranging from 30 in. Hg down to 8.8 in. Hg (approximately 30,000 ft Altitude), the pressure remaining constant for long periods of time or varying at a rate as high as 0.5 in. Hg per second.

(2) Non-operating.- As specified for class 1 equipment in Specification MIL-E-5400.

3.4.7.3 Humidity.- (See 4.11.2) As specified in Specification MIL-E-5400.

3.4.7.4 Vibration.- (See 4.11.3)

a. Transmitter-receiver on mount.-

(1) Non-operating.- The equipment shall meet the requirements of Specification MIL-E-5400 pertaining to vibration under storage and transit conditions, except that tests for compliance with the paragraph of Specification MIL-E-5400 titled "Equipment with vibration isolators removed" shall be made in the vertical plane only.

(2) Operation and standby.- For operation and standby conditions (operating and non-operating) the equipment shall be tested in the normal installation position or orientation (with vibration isolators in place).

b. Transmitter-receiver without mount.-

(1) Non-operating.- In accordance with requirements specified in 3.4.7.4a(1).

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3.4.7.4 Vibration.- (Cont'd)

(2) Operation and standby.- In accordance with requirements specified in 3.4.7.4a(2).

c. Control unit.- (See 4.11.3.1b) The control unit shall meet the vibration requirements of Specification MIL-E-5400.

d. Antenna.- (See 4.11.3.4) The antenna shall meet the vibration requirements of Specification MIL-E-5400.

3.4.7.5 Shock.- See 4.11.4) The equipment shall meet the shock requirements of Specification MIL-E-5400.

3.4.7.6 Explosion conditions.- (See 4.11.5) The equipment shall meet the explosion requirements of Specification MIL-E-5400.

3.4.7.7 Rain:- (See 4.11.6) A simulated rainfall of 4 ± 1 inch per hour.

3.4.7.8 Sand and dust.- (See 4.11.7) As specified in MIL-E-5400.

3.5 Weight.- The total weight of the equipment listed in 1.1 shall be as follows:

3.5.1 Receiver-Transmitter RT-494( )/APX-44 with Mounting MT-2100( )/APX-44.- The overall weight of the receiver-transmitter, associated power supply including connectors and mounting base, but not including any external cabling shall not exceed 25 pounds.

3.5.2 Control unit.- The weight of this unit shall not exceed 2.75 pounds.

3.5.3 Antenna unit.- The weight of this unit shall not exceed 0.75 pound.

3.6 Input power requirements.- The Transponder Set shall meet all applicable requirements of Specification MIL-E-7894 and shall give full specified performance when operated from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein:

a. DC power 27.5 volts (system), at 22 watts maximum.

b. 25 to 29 volts D.C. (operating voltage limits).

c. 20 to 25 volts D.C. and 29 to 30 volts D.C. without damage to the unit components or circuitry.

3.6.1 Undervoltage protection.- The equipment shall incorporate circuitry so that no damage will occur by voltages below 20v D.C. and shall automatically resume normal operation when the voltage returns within operating limits.

3.6.2 Overload protection.- Overload protection shall be provided by means of fuses or circuit breakers located so they are accessible from the front of the unit.

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3.7 Performance.- The Transponder Set shall meet the performance characteristics specified in 3.7.1 through 3.13.8.4c inclusive.

3.7.1 Conditions for performance.- Unless otherwise stated, performance requirements defined herein are based on: (See 6.4 through 6.4.15).

a. Maximum warm-up time shall not exceed five (5) minutes.

b. Interrogation signals having characteristics as described in 3.11.1.

c. Interrogation rates from 15 to 500 interrogations per second.

d. Service conditions as specified in 3.4.7 (operating) of this specification.

e. Pulse dimensions and characteristics as specified in 3.13.6.

3.8 Receiver-Transmitter RT-494( )/APX-44.- The receiver-transmitter unit shall be constructed to meet the requirements of 3.8.1 through 3.8.4.2 inclusive.

3.8.1 Function.- The receiver-transmitter shall receive interrogations at the antenna receptacle and transmit coded replies as a function of the control unit settings.

3.8.2 Form factor.- The receiver-transmitter shall be contained in a one-half ATR unit as per the procurement model.

3.8.3 Contents of unit.- The Receiver-Transmitter RT-494( )/APX-44( ) shall contain the following items:

a. Superheterodyne receiver including Local Oscillator, Preselector, Crystal Mixer, I.F. Amplifier and Detector.

b. Video amplifier containing Spike discriminator, and Echo suppression circuitry.

c. Modes 1,2 and 3 Decoders.

d. Pulse Train Coder.

e. Multi-Pulse and Reply Pulse Generator.

f. Transmitter including R.F. Oscillator, Modulator, and Pulse Generator.

g. Power supply.

h. Control relays, Control circuits and receptacles.

3.8.4 Crystal oscillator.-

3.8.4.1 Power dissipation of crystal unit.- The allowable power dissipation of the oscillator crystal unit shall not exceed 17.0 milliwatts.

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3.8.4.2 Frequency of crystal unit.- The allowable upper frequency limit of the crystal unit shall be 91 mc/s.

3.9 Control, Transponder Set C-2714( )/APX-44.- The control unit shall be constructed to meet the requirements specified in 3.9.1 through 3.9.3 inclusive.

3.9.1 Function.- This unit shall provide for the remote control of certain facilities in Receiver-Transmitter RT-494( )/APX-44 as specified herein.

3.9.2 Controls.- The following controls shall be provided to accomplish the function described herein:

a. Master.- This shall be a four position selector switch with positions "OFF", "LOW", "NORM" and "EMER". When switched to the "LOW" position, the sensitivity of the transponder shall be reduced by approximately 30 db, adjustable by a potentiometer within the equipment. (See 3.12.2). Means shall be provided to prevent accidental switchings to "EMER".

b. Mode controls.- This shall consist of two single-pole single-throw toggle switches labelled "Mode 2" and "Mode 3" and "ON" - "OFF".

c. Code controls.- This shall consist of two coaxial type switches labelled "Mode 1" and "Mode 3". The code numbers on the dials shall conform to the code numbering assignment outlined in 3.13.8.4.

d. I/P.- This shall consist of a three position toggle switch. The up position shall be labelled "I/P", center position labelled "OFF" and down position labelled "MIC". The up "I/P" position shall have spring-loaded momentary type operation.

e. Function control.- This shall consist of a three position switch marked "NORMAL", "MOD", and "CIVIL".

f. Monitor control.- This shall consist of a single-pole single-throw switch marked "ON", and "OFF".

3.9.3 Lighting plate.- The plastic lighting plate shall meet the requirements of MIL-P-7788.

3.10 Antenna AT-884( )/APX-44.- The antenna shall be constructed to receive vertically polarized signals at 1010-1030 mc/s and radiate vertically polarized signals at 1090-1110 mc/s. The antenna shall be omni-directional and shall be constructed so that the transponder is capable of performing its function during all normal maneuvers of the aircraft at all azimuthal headings with respect to any radial to the interrogating radar location.

3.10.1 Antenna protection.- The words "DO NOT PAINT" shall appear in a suitable place on the antenna.

3.10.2 Antenna flange.- The antenna shall be equipped with a suitable flange to permit affixing of an antenna test probe such as Prod Test MX-1342/UPM-4A.

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3.10.3 Antenna mismatch.- The antenna system shall match to a 50-ohm coaxial cable and shall present a mismatch not greater than 3 db for all combinations of service conditions (operating) of 3.4.7.2 and 3.4.7.3.

3.11 Overall operation.- The equipment shall receive coded interrogations, shall decode these interrogations, and, in response, shall transmit pulse replies.

3.11.1 Interrogation characteristics.- The equipment shall be constructed for interrogation signals consisting of pairs of pulses. The interrogation pulse will have a duration within the limits of 0.7 and 1.2 microseconds, a rise time not greater than 0.2 usec, and a decay time of not greater than 0.4 microseconds. The variation in amplitude between pulses of an interrogation shall not exceed 3 db. The pulse separation of an interrogation will constitute the interrogation "Mode" as follows:

Mode	Pulse separation
1	3.0 ± 0.2 microseconds
2	5.0 ± 0.2 microseconds
3	8.0 ± 0.2 microseconds

3.11.2 Frequency characteristics.- The equipment shall be constructed to operate in the 960 to 1215 mc. band assigned to air navigation services and have minimum tuning capability as follows:

Receiver tuning range	Transmitter tuning range
1010 to 1030 mc./s	1090 to 1110 mc/s

3.11.3 Receiver protection.- The receiver and all other assemblies of the Receiver-Transmitter unit shall not be damaged by the transmitter power of this equipment.

3.11.4 Receiver frequency drift.- After a 3-minute warm-up the total frequency drift of the entire receiver over a 24-hour period shall not exceed ± 1.0 mc. under all reasonable combinations of service conditions (operating) specified in 3.4.7.

3.11.5 Transmitter frequency drift.- The frequency drift of the transmitter shall not exceed ± 3 mc/s, when tested in accordance with 4.9.2, under the following conditions:

a. Under all service conditions (operating) of 3.4.7.

b. Mismatching at the antenna connection of the transponder which produces a voltage standing wave ratio not exceeding 3.5 db.

c. Variations of interrogation rates of 500 to 2000 interrogations per second.

3.12 Receiving characteristics.-

3.12.1 Triggering level.- The normal triggering level at maximum response, in any pass band within the operating frequency range of the equipment, shall be between 72 and 76 db below one milliwatt. (See 4.9.9 and 6.4.15).

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3.12.2 Triggering level selection.-Provision shall be incorporated within the equipment for remote selection of two triggering levels as follows:

(a) Normal triggering level - as defined in 3.12.1.

(b) Low triggering - selected by "Low" switch on Control Transponder Set C-2714( )/APX-44 to any preset value between the normal triggering level and -42 dbm.

3.12.3 Triggering level drift.- The triggering level (when set to a nominal service value) shall not vary by more than 3 db from the original nominal value, over all reasonable combinations of service conditions with the exception that standard operating conditions of voltage shall apply. The triggering level shall not vary by more than 6 db (from the original nominal value), over all reasonable combinations of service conditions.

3.12.4 Random triggering rate.- The random triggering rate shall not exceed 5 replies per second in the average as integrated over an interval equivalent to at least 30 seconds.

3.12.5 Overall receiver triggering bandwidth.- The triggering bandwidth of the receiver 3 db from maximum response shall be 6 to 8 mc/s. The triggering bandwidth shall not be more than 28 mc/s at 40 db down from maximum response, making suitable correction for the presence of frequency modulation in the output of the signal generator employed.

3.12.6 Bandwidth drift.- The receiver shall not lose nor gain more than 3 mc/s in overall bandwidth for any of the operating service conditions specified in 3.4.7 over a period of 6 hours. No artificial heating systems nor thermostatically controlled temperature stabilizing devices shall be used to maintain receiver frequency stability.

3.12.7 I. F. Frequency and bandwidth.-

3.12.7.1 I. F. Frequency.- The I. F. mid-frequency between points 3 db down from maximum response shall be 60 ± 1 mc.

3.12.7.2 I. F. Bandwidth.- The I. F. bandwidth shall be at least 6 mc/s at 3 db points.

3.12.7.3 I. F. Pick-up.- The intermediate frequency amplifier shall be substantially shielded from extraneous, radiated energy pick-up at any frequency in the IF band. The triggering level for 60 mc/s energy on the antenna terminal of the equipment shall be not less than 42 db down from the normal triggering level at the tuned frequency of the receiver.

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3.12.8 Image response.- The image response measured under normal triggering level conditions shall be such that the maximum response to signals of image frequencies shall be at least 50 db down from the normal triggering level at the tuned frequency of the receiver.

3.12.9 R.F. rejection.- RF rejection shall be provided which will reduce the triggering level to the following frequency bands by a minimum of 60 db below the triggering level at the frequency to which the receiver is tuned.

(a) 0.1 to 1000 mc/s

(b) 1840 to 2360 mc/s

(c) 2790 to 3510 mc/s

(d) 8490 to 12160 mc/s

3.12.10 Anti-jamming.- The equipment shall operate in the presence of simple CW, AMCW, random-spaced pulses, and regularly-spaced pulses of any spacing as follows:

(a) If the interrogating signal to which the equipment should reply is between 3 and 6 db above the normal triggering level, an interfering signal 10 db down from the interrogating signal shall not reduce the number of replies by more than 50 per cent.

(b) If the interrogating signal is between 20 and 30 db above the normal triggering level, an interfering signal at a level 3 db down from the interrogating signal shall not reduce the number of replies by more than 50 per cent.

3.12.11 Interrogation rates.- The equipment shall be constructed for operation with normal interrogation rates of from 15 to 500 interrogations per second. However, the duty factor capability of the equipment shall be such as to permit operation with interrogation rates as high as 2000 interrogations per second with an 8-pulse reply code. Circuitry shall be provided to protect the equipment from damage due to over interrogation. Additionally, to protect the overall system from the effects of transponder over interrogation as the result of side and backlobe radiation from ground interrogation antennas by denying service to weaker signals when a predetermined maximum has been reached, a sensitivity-reduction type reply limit control shall be incorporated in the equipment. The threshold or turndown point of this circuitry shall be adjustable between 500 and 2000 interrogations per second and shall be more or less independent of the number of pulses in the reply. When this over interrogation control has been set to a predetermined value, the transponder triggering level shall not be reduced by more than 3 db for interrogation rates approaching that value and shall be reduced by more than 30 db for interrogation rates in excess of 50 to 100% greater than the predetermined value. The second, third and fourth intermediate frequency amplifier stages shall have automatic overload control.

3.12.12 Other spurious responses.- All other spurious responses measured under normal triggering conditions shall be such that the maximum response to such signals shall be at least 60 db down from the normal triggering level of the receiver. An external filter may be used, if necessary, in order to meet this requirement. When filters are considered necessary, they shall be provided with the equipment.

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3.12.13 Decoder.- The equipment shall incorporate decoding circuits for the purpose of distinguishing the three modes of interrogation described in 3.11.1. The following requirements shall apply when interrogating signals described in 3.11.1 at levels from 3 to 40 db above the triggering level are applied.

3.12.13.1 Decoder tolerances - Mode 1.- When set to Mode 1, the equipment:

a. Shall transmit a reply in response to a Mode 1 interrogation, and

b. Shall not respond to double-pulse interrogations when the pulses are spaced equal to or less than 2.0 microseconds, or equal to or greater than 4.0 microseconds.

3.12.13.2 Decoder tolerances - Mode 2.- When set to Mode 2, the equipment:

a. Shall transmit a Mode 1 reply to a Mode 1 interrogation, and a Mode 2 reply to a Mode 2 interrogation.

b. Shall not respond with Mode 2 replies to double-pulsed interrogations when the pulses are spaced equal to or less than 4.0 or equal to or greater than 6.0 microseconds; and

c. Shall not respond with Mode 1 replies to double-pulsed interrogations when the pulses are spaced equal to or less than 2.0 microseconds, or equal to or greater than 4.0 microseconds.

3.12.13.3 Decoder tolerances - Mode 3.- When set to Mode 3, the equipment:

a. Shall transmit a reply to both Modes 1 and 3 interrogations; and

b. Shall not respond to double-pulse interrogations when the pulses are spaced equal to or less than 2.0 microseconds; or between 4.0 and 7.0 microseconds inclusive, or equal to or greater than 9.0 microseconds.

3.12.13.4 Decoder tolerances - Modes 2 and 3.- When set for simultaneous operations of Modes 2 and 3, the equipment:

a. Shall transmit Modes 1, 2, and 3 replies to Modes 1, 2, and 3 interrogations;

b. Shall not transmit a reply in response to double-pulse interrogations when the pulses are spaced between 6.0 and 7.0 microseconds; and

c. Shall not respond to double-pulse interrogations when the pulses are spaced equal to or less than 2.0 or equal to or greater than 9.0 microseconds.

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3.12.13.5 Decoder - single pulse.- The equipment shall not respond to a single pulse interrogation signal of any duration, provided that;

a. The pulse carries no ripples comparable to normal interrogations;

b. No CW capable of producing beats comparable to normal interrogation is present; and

c. The single pulse is either in tune or if not in tune, is not of a duration corresponding to a normal interrogation pulse separation. Particular care shall be taken to minimize response to pulses of the TACAN signal.

3.12.13.6 Decoder control.- The equipment shall be so constructed that the receiver-transmitter and the control unit will permit selection of any one of the following conditions of operation:

a. Mode 1

b. Modes 1 and 2

c. Modes 1, and 3

d. Modes 1, 2 and 3

3.12.14 Echo effects.- The equipment shall operate in the presence of echoes of the interrogating signals in accordance with the following:

a. When the signal strength of the interrogating signal to which the equipment should reply is between 3 and 10 db above the normal triggering level, an echo of any duration, with its leading edge lying between the two pulses of the interrogation, shall not reduce the number of replies by more than 50 per cent provided the signal strength of the interrogation exceeds that of the echo by at least 10 db.

b. When the signal strength of the interrogation is between 20 and 40 db above the normal triggering level, an echo of any duration, with its leading edge lying between pulses of the interrogation, shall not reduce the number of replies by more than 50 percent provided that the signal strength of the interrogation exceeds that of the echo by at least 6 db.

c. For test purposes, the echo may be simulated by a signal of slightly different frequency (but not greater than one megacycle difference) than the interrogation.

d. With the equipment set to Mode 1, replies shall not be transmitted in response to the reception of a single interrogation followed by an echo with the echo spaced for most probable triggering provided that the signal strength of the single pulse does not exceed the normal triggering level more than 50 db and exceeds that of the echo by 35 db.

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3.12.14 Echo effects.- (Cont'd)

e. The equipment shall regain its sensitivity within 3 db of the minimum triggering level within 20 microseconds after reception of a 1.0 microsecond pulse having a signal strength of 50 db above the minimum triggering level.

f. When a 1.0 microsecond reference pulse with an amplitude of less than -22 dbm is followed at 2 microsecond (leading edge to leading edge) by a 0.5 microsecond control pulse at least 10-db below the 1 microsecond pulse, the 0.5 microsecond control pulse shall be reduced in amplitude at the video output jack. In a like manner, when a series of four 0.5 microsecond reference pulses, all spaced 2 microseconds apart is followed at two microseconds by a 0.5 microsecond control pulse, at least 10 db below the referenced four (4) pulse train, the 0.5 microsecond control pulse shall be reduced in amplitude at the output jack. This reduction in amplitude shall be such that the control pulse video amplitude at the receiver output receptacle will be less than that caused by a 0.5 microsecond r-f pulse at 6 db below the transponder minimum triggering level.

3.12.15 Echo effects - decoder.- The equipment shall meet the following requirements when in the presence of echoes as specified in 3.12.14.

3.12.15.1 Decoder - Mode 1 with echoes present.- When set to Mode 1, the equipment shall transmit a Mode 1 reply in response to a Mode 1 interrogation.

3.12.15.2 Decoder - Mode 2, with echoes present.- When set to Mode 2, the equipment:

a. Shall transmit a Mode 1 reply in response to a Mode 1 interrogation

b. Shall transmit a Mode 2 or Mode 1 reply in response to a Mode 2 interrogation depending upon the position of the echo with respect to the interrogation; and

c. Shall not transmit a Mode 2 reply in response to a Mode 1 interrogation.

3.12.15.3 Decoder - Mode 3, with echoes present.- When set for Mode 3, the equipment shall transmit a Mode 1 reply in response to both Mode 1 and Mode 3 interrogations.

3.12.15.4 Decoder - Modes 2 and 3, with echoes present.- When set for simultaneous operation of Modes 2 and 3, the equipment:

a. Shall transmit a Mode 1 reply in response to a Mode 1 interrogation;

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3.12.15.4 Decoder-Modes 2 and 3, with echoes present.- (Cont'd)

b. Shall transmit a Mode 2 or Mode 1 reply in response to Mode 2 or Mode 3 interrogation, depedning upon the position of the echo; and

c. Shall not transmit a Mode 2 reply in response to a Mode 1 interrogation.

3.12.15.5 Decoder - emergency reply, with echoes present.- When set to emergency the transponder shall reply in response to Modes 1, 2, and 3 interrogations.

3.12.16 Suppression.-

3.12.16.1 Suppression receptacle.- An external suppression, input-output, receptacle shall be provided on the front panel. The receptacle shall provide for connection to an external suppression pulse generator capable of producing suppression pulses having the characteristics of 3.12.16.2 and shall provide for output of suppression pulses from the equipment.

3.12.16.2 Characteristics.- The equipment shall be constructed for suppression based upon input-output suppression pulses having the following characteristics:

a. Amplitude - 20 to 70 volts

b. Polarity - positive

c. Duration - 1.5 to 250 microseconds

d. Output impedance - 5000 ohms (minimum)

or

Input impedance - 2000 ohms (minimum)
20 micro-microfarads (maximum)

e. Rise time - equal to or greater than 10 volts per microsecond

f. Decay time - equal to or greater than 10 volts per microsecond

3.12.16.3 Operation.- The receiver shall be rendered as close to completely insensitive as practicable, with the application of a suppression pulse having the characteristics of 3.12.6.2, and in any event the triggering level shall be reduced to less than 14 db below one volt for an interval starting not more than one microsecond after the suppression pulse has reached 10 volts amplitude and lasting until the suppression pulse has fallen to an amplitude of 10 volts. The triggering level shall have recovered to within 3 db of normal between 15 and 125 microseconds after the termination of the suppression pulse.

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3.12.17 Radar interference.- The reply rate of the equipment due to a normal interrogation shall not be altered if the interrogation signal is preceded by a single two-microsecond pulse provided:

(a) The signal strength of the single pulse measured at the antenna terminal of the receiver-transmitter does not exceed that of a similar pulse whose frequency is that to which the receiver is tuned, and

(b) That the signal strength of the pulse measured at the antenna terminal is less than that of the interrogation signal by 10 db.

3.12.17.1 Conditions.- This requirement shall hold for interrogation signal strengths from 20 to 50 db above the triggering level and with the radar pulse spaced from the interrogation by an amount which is most likely to cause unwanted triggering. It is assumed that under these conditions the reply may be advanced in time equal to the separation between the radar pulse and the first of the normal interrogation pulses.

3.12.18 Transponder suppressed time.- After reception of a proper interrogation, the transmitter shall not reply to any other interrogation for the duration of the reply pulse train. The suppressed time shall end no earlier than 75 microsecond and no later than 150 microsecond after transmission of the last reply pulse of a normal SIF group. The suppressed time desensitization shall be such that the receiver is at least 30 db less sensitive than the normal triggering level at the 50 microsecond point and that it is recovered to within 3 db of the normal triggering level at the 150 microsecond point.

3.12.19 Spike suppression.- The equipment shall discriminate against all pulses received in the equipment which have less than 0.3 microseconds duration in order to reduce triggering due to extraneous noise spikes and due to narrow pulses which might be generated by external equipment, pulses not in tune, or noise.

3.12.20 Receiver video output.- A front panel receptacle shall be provided as an outlet for receiver video signals. This output shall have the following characteristics:

a. Amplitude.- Not less than 1.0 volt when the input signal level to the receiver is -74 dbm.

b. Polarity.- Positive

c. Output impedance.- Not less than 1000 ohms with not more than 700 uuf capacitance to ground.

d. Source impedance.- Not more than 100 ohms.

e. Droop.- The droop experienced by a pulse train consisting of forty 0.5 microsecond pulses spaced 2 microseconds apart, shall be less than 1 db. The droop shall be measured by taking the peak-to-peak value of the first 0.5 microsecond pulse and comparing it to the peak-to-peak value of the last pulse in the train.

f. Narrowing.- The last pulse of this train of 0.5 microsecond pulses, spaced 2.0 microseconds apart at a level less than -22 dbm shall be

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3.12.20 Receiver video output.- (Cont'd)

f. Narrowing.- (Cont'd) narrowed more than 0.2 microseconds.

g. The second pulse of a Mode 1 interrogation at a level less than -22 dbm shall not be narrowed more than 0.2 microseconds (measured at one-half amplitude at the r-f pulse and at the threshold decoding level of the video signal at the receiver output).

3.13 Transmitter characteristics.-

3.13.1 Power output.- With the equipment set for Mode 1, and with an interrogation rate of 500 per second, the transmitter shall deliver as close to 27 db above one watt as practicable, and in no event less than 24 db nor more than 30 db above one watt RF power, (as averaged over an RF pulse) to a 51 ohm resistive termination at the antenna terminal of the equipment at 500 pulses per second. The power output shall not be reduced to less than 22 db above one watt due to load impedance mismatch up to 3.0 db (1.4:1 VSWR). Independent of the above listed variations in power output, the power output shall be reduced by not more than one db when the Mode 1 reply rate is increased from 500 to 2000 per second, and by not more than two db additional when further increased to 3500 per second. The average pulse power output of the equipment for Modes 2 and 3 and for emergency shall be substantially equal to Mode 1 as defined above.

3.13.1.1 Transmitter load protection.- There shall be no damage to the transponder transmitter if the antenna or transmission line is accidentally opened or shorted.

3.13.2 Characteristics of reply.- The equipment shall respond to interrogations defined in 3.11.1 by transmitting replies consisting of 1, 2, or 4 pulse or pulse trains for each interrogation received depending on the mode of interrogation used and upon the equipment control setting selected. The equipment shall also provide optional use of Coded Pulse Train Replies as specified in the paragraphs that follows. Availability of this type of operation shall be provided by an externally located selector switch. When the switch is in a "Normal" position, the equipment shall provide reply pulses as described in 3.13.2.1 to 3.13.3.2 and 3.13.6.2 of this specification. When the switch is in the "Modified" position the equipment shall produce coded pulse train replies as described in 3.13.4 to 3.13.5 and 3.13.6.2.

3.13.2.1 Mode 1 reply.- A Mode 1 reply shall consist of one pulse having dimensions specified in 3.13.6.1.

3.13.2.2 Mode 2 reply.- A mode 2 reply shall consist of two pulses each having dimensions specified in 3.13.6.1 and spaced 15.95 ± 0.1 microseconds apart.

3.13.2.3 Mode 3 reply.- Same as Mode 1 reply, as described in 3.13.2.1.

3.13.2.4 Emergency reply.- An emergency reply shall consist of four pulses having dimensions specified in 3.13.6.1 and spaced 16 ±2.5 microseconds apart. The emergency reply may be made in either Mode 1, 2 or 3.

3.13.3 I/P provisions.- The equipment shall incorporate two methods of providing I/P (identification of position) operation controlled by the "I/P-OFF-MIC" switch specified in paragraph 3.9.2d.

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3.13.3.1 "I/P" position.- When the switch described in 3.9.2e is in the "NORMAL" position and the "I/P" position of the switch specified in 3.9.2d is selected, a Mode 2 reply shall be transmitted for each interrogation received regardless of mode. Upon release of the "I/P" switch the operation of the equipment shall revert to the designated mode of operation after normal "I/P" delay described in 3.13.5.2.1.

3.13.3.2 "Mic" position.- When the switch described in 3.9.2e is in the "NORMAL" position and the "Mic" position of the switch specified in 3.9.2d is selected, the circuitry shall be so arranged as to control the I/P function from an "ON-OFF" type switch external to the equipment. When the external switch is in the "ON" position, the equipment shall operate as specified in 3.13.3.1.

3.13.4 Coded pulse train reply.-

3.13.4.1 Mode 1 - Coded pulse train reply.- For each Mode 1 interrogation a pulse train shall be generated containing none to a maximum of five information pulses plus two framing pulses which are spaced 20.3 ±0.05 microseconds. The spacing between pulses shall be 2.9 ±0.05 microseconds and shall be measured in 2.9 microsecond multiples from the initial framing pulse. The position where a sixth information pulse would appear, i.e. 17.4 ±0.05 microseconds from the initial framing pulse, will not be used. The condition wherein none or all five information pulses are present will not be employed operationally. From the specified five information pulses, a total of thirty-two different codes remotely selectable shall be available.

3.13.4.2 Mode 2 - Code pulse train reply.- For each Mode 2 interrogation, a pulse train shall be generated containing none to a maximum of twelve information pulses plus two framing pulses which are spaced 20.3 ±0.05 microseconds. The spacing between pulses shall be 1.45 ±0.05 microsecond multiples from the initial framing pulse. The position where the seventh information pulse would appear, i.e. 10.15 ±0.05 microseconds from the initial framing pulse, will not be used. Facility for selecting any code combination of the information pulses specified shall be provided on the front panel of the receiver-transmitter by switches (at least one switch shall be marked to indicate "ON-OFF" positions).

3.13.4.3 Mode 3 - Coded pulse train reply.- For each Mode 3 interrogation, a pulse train shall be generated containing none to a maximum of six information pulses plus two framing pulses which are spaced 20.3 ±0.05 microseconds. The spacing between pulses shall be 2.9 ±0.05 microseconds and shall be measured in 2.9 microsecond multiples from the initial framing pulse. The condition wherein none or all six information pulses are present will not be employed operationally. Facility for remotely selecting any of the 64 code combinations of the information pulses specified shall be provided.

3.13.5 Emergency and "I/P" operation (SIF).-

3.13.5.1 Emergency operation.- When the switch described in 3.9.2e is in "MOD" position and the selector switch on the Control Unit is switched to the emergency position, the Mode 1 reply shall be the Mode 1 pulse train (code) repeated four times for each interrogation received and spaced 4.35 ± 0.1 microseconds apart. The Mode 2 reply shall be the single normal Mode 2 pulse train (code) reply. The Mode 3 reply shall be the Mode 3 pulse train (code) repeated four times for each interrogation received and spaced 4.35 ±0.1 microseconds apart. The pulse dimensions shall be as specified in 3.13.6.2.

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3.13.5.2 "I/P" operation.- The equipment shall incprporate two methods of providing I/P (identification of position) operation by the "I/P" - "OFF" - "MIC" switch specified in 3.9.2d.

3.13.5.2.1 "I/P" position.- When the switch described in 3.9.2e is in the "MOD" position and the "I/P" position of the switch specified in 3.9.2d is selected, the Mode 1 reply shall be a Mode 1 pulse train (code) repeated twice for each interrogation received and speced 4.35 ±0.1 microseconds apart. The Mode 3 reply shall be the normal Mode 3 pulse train (code) repeated twice for each interrogation received and speced 4.35 ±0.1 microseconds apart. The Mode 2 reply is the Normal Single Mode 2 code reply. The pulse dimensions shall be as specified in 3.13.6.2. The "I/P" circuitry shall be so arranged that "I/P" operation shall continue 30 $^{+10}_{-20}$ microseconds after the "I/P" switch is released.

3.13.5.3 "MIC" position.- When the switch described in 3.9.2e is in the "MOD" position and the "MIC" position of the switch specified in 3.9.2d is selected, the circuitry shall be so arranged as to control the I/P function by an "ON-OFF" type switch external to the equipment. When the external switch is in the "ON" position, the equipment shall operate as specified in 3.13.5.2.1.

3.13.6 Pulse dimensions and characteristics.-

3.13.6.1 "NORMAL" operation.- When the equipment is set for "NORMAL" operation, the reply pulse or pulses shall have the following characteristics:

(a) Duration - 0.45 ±0.1 microseconds.

(b) Rise time - 0.1 microsecond, or less.

(c) Decay time - 0.2 microseconds or less.

(d) Amplitude variation - 10 percent, or less.

(e) Spike amplitude - not greater than 40% of the average pulse amplitude.

(f) Amplitude jitter - 5% or less.

3.13.6.2 Modified operation (SIF).- When the equipment is set for "MOD" operation, the pulse or pulses of the reply train shall have the following characteristics:

a. Duration - 0.45 ±0.1 microsecond.

b. Rise time - 0.1 microsecond, or less.

c. Decay time - 0.2 microsecond, or less.

d. Amplitude variation - 10% or less.

e. Spike variation - Not greater than 40% of the average pulse amplitude.

f. Amplitude jitter - 5% or less.

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3.13.6.2.1 Special reply pulse.- The pulse width of the transmitted R.F. pulse shall not be less than 0.40 microseconds when the transponder is externally triggered at the trigger-in receptacle, located on the front panel of the receiver-transmitter, by pulse-paired trigger pulses spaced 4.0 microseconds apart and at 1000 pulse-pairs per second.

3.13.7 Civil identification.-

3.13.7.1 "I/P" operation.- When the switch described in 3.9.2e is in the "CIVIL" position and the "I/P" position of the switch specified in 3.9.2d is selected, and the transponder is interrogated in Mode 3, the reply shall be a single pulse spaced 4.35 ±0.1 microseconds after the last pulse (framing pulse) in the Mode 3 code train.

3.13.7.2 Other replys.- All other functions of the transponder shall be identical to those specified under normal operation.

3.13.8 Transponder characteristics.-

3.13.8.1 Transponder delay.- The transponder delay shall not exceed 3.0 ±0.3 microseconds measured when the received pulses have exactly the nominal spacing and have an amplitude varying from 6 to 50 db above the triggering level.

3.13.8.2 Range jitter.- The range jitter on successive replys shall not exceed 0.2 microseconds.

3.13.8.3 Spurious responses.- Transmitter spurious responses shall be down 60 db from the pulse amplitude of the on-frequency transmitted pulse.

3.13.8.4 Code number assignments.- Every configuration of information pulses on each mode shall be assigned a code number. The following code numbers shall be used:

a. Mode 1.- Code numbers 00 through 73 shall designate the Mode 1 code. The applicable five information pulses are divided into two groups, A and B, A containing the first three pulses and B containing the last two pulses. The three information pulses in Group A are numbered in order 1, 2, and 4 and the two pulses in Group B are numbered in order 1, and 2. The appropriate code number is obtained by adding the pulse numbers (1, 2 and 4) of the pulses present within each group and combining the two sums in the order AB to form the serial code number. Thus 03 consists of no pulses in Group A and pulses 1 and 2 of Group B, code 51 consists of pulses 1 and 4 of Group A and pulse 1 of Group B, etc.

b. Mode 2.- Code numbers 0000 through 7777 shall designate the Mode 2 code. The applicable twelve information pulses are divided into four groups of three pulses, the pulses of each group being separated by 2.9 microseconds and numbered in order 1, 2 and 4. The position of the first pulse of each group from the first framing pulse is as follows:

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3.13.8.4 Code number assignments.- (Cont'd)

b. Mode 2.- (Contd)

Group	Time From First Framing Pulse (Microsecond)
A	2.9
B	11.60
C	1.45
D	13.05

The appropriate code number is determined by adding the pulse numbers (1, 2, and 4) of the pulses present within each group. These sums are then combined in the order A, B, C, D to form a 4 digit serial number for the particular code. For example, Code 2435 consists of pulse 2 of Group A, pulse 4 of Group B, pulses 1 and 2 of Group C and pulses 1 and 4 of Group D. The particular pulse train for code 2435 will therefore have information pulses at the first, third, fourth, ninth, twelfth and thirteenth positions.

c. Mode 3.- Code numbers 00 through 77 shall designate the Mode 3 code. The same system applicable to Mode 1 as described in 3.13.8.4a is employed except that the sixth information pulse (pulse number 4 of Group B) is included.

3.14 Equipment life, accelerated.- (operating) The equipment will be subjected to continuous use for long periods under varied and severe military service conditions without overhaul and with little, if any, maintenance. To meet these conditions, the equipment shall have a survival rate of 90 percent at the 24 hour period and 81 percent at the 48 hour period as will be indicated by no more than 7 failures in 5 test units when tested as specified in 4.14.

3.15 Preconditioning.- Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.5. (Also see 4.4d)

3.16 Technical literature, tools, and running spare parts.- Technical literature, all tools required for installation and operation including special tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.17 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract.

3.18 Workmanship.- The equipment shall be manufactured and assembled in accordance with the workmanship requirements of Specification MIL-E-5400, and the workmanship requirements of this specification as listed below:

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3.18 Workmanship.- (Cont'd)

Paragraph	Requirement
	Printed wiring board:-
3.4.5.1a	Etching.
	Electronic part mounting:-
3.4.5.2a	Lead-mounted parts.
3.4.5.2b	Axial lead shaping.
3.4.5.2f	Leads, after insertion.
3.4.5.2g	Soldered joint reheating.
	Processing:-
3.4.5.3a	Soldering.
3.4.5.3b	Flux.
3.4.5.4a(1)	Thickness of protective coating on printed wiring boards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility.- Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (See 4.3).

(b) Procurement inspection (shall be the inspection performed by the contractor and the Government, as specified by 4.4 and 4.15.

(1) Procurement inspection of items before preparation for delivery. (See 4.4).

(2) Procurement inspection for delivery. (See 4.15 and section 5).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and the inspection specified for group A, group B, and group C (see tables II, III and IV, respectively.) The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, and (3) Explosion-condition; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

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Table I - Preproduction inspection

Inspection	Rqt Par	Test Par
Protective Coating Tests (printed circuit boards)	3.4.5.4	4.13
Temperature-altitude	3.4.7.2	4.11.1
Vibration	3.4.7.4	4.11.3
Shock	3.4.7.5	4.11.4
Humidity	3.4.7.3	4.11.2
Explosion-condition	3.4.7.6	4.11.5
Sand and dust	3.4.7.8	4.11.7
Rain (antenna only)	3.4.7.7	4.11.6

4.4 Procurement inspection of items before preparation for delivery.- The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection.)

a. Procurement inspection shall consist of group A, group B, and group C inspection as specified in 4.4.1 through 4.4.3.3.

b. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

c. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

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4.4 Procurement inspection of items before preparation for delivery.- (Cont'd)

d. Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.15).

4.4.1 Group A inspection.- Group A inspection (including sampling) shall conform to Table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in the order which is satisfactory to the Government inspector.

Table II - Group A inspection

Inspection	Rqt Par	Test Par	AQL Major	AQL Minor
Visual and Mechanical		4.6		
Receiver-Transmitter RT-494( )/APX-44	-	(4.6.1) (4.6.2)	2.5dphu	10 dphu
Control, Transponder C-2714( )/APX-44	-	4.6.1	1.0%	4.0%
Antenna AT-884( )/APX-44	-	4.6.1	1.0%	4.0%
Mount MT-2100( )/APX-44	-	4.6.1	1.0%	4.0%
Electrical				
Printed circuit boards assembled				
Dielectric strength	3.4.5.4b(2)	4.13.6.1	1%	
Receiver-Transmitter RT-494( )/APX-44 and Control, Transponder C-2714( )/44				
Transponder				
Delay Range Pulse reply	3.13.8.1 3.13.8.2 3.13.2 through 3.13.7.2	4.9.4 4.9.27 4.9.6	4.0 dphu for the group	
Code number assignment	(3.13.8.4a,b,c)	4.9.6.2		
Triggering Level	3.12.1	4.9.9		
Triggering Level Selection	3.12.2	4.9.9.1		
Triggering Bandwidth	3.12.5	4.9.12.2		
Transmitter Power Output	3.13.1	4.9.3		

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Table II - Group A inspection (Cont'd)

Inspection	Rqt Par	Test Par	AQL Major	AQL Minor
Transponder (Cont'd)				
Decoder	3.12.13	4.9.19		
Decoder Tolerance Mode - 1	3.12.13.1	4.9.19.1		
Decoder Tolerance Mode - 2	3.12.13.2	4.9.19.2		
Decoder Tolerance Mode - 3	3.12.13.3	4.9.19.3		
Decoder Tolerance Modes - 2 and 3	3.12.13.4	4.9.19.4		
Decoder single pulse	3.12.13.5	4.9.19.5		
Decoder control	3.12.13.6	4.9.19.6		
Suppression receptacle	3.12.16.1	4.9.21.2		

4.4.2 Group B inspection.- This inspection (including sampling) shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.4.2.1 Group B sampling plans.- The group B sampling plans, for the AQL listed in table III, shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
6.5%	L-7	L-5

4.4.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B inspection

Inspection	Rqt Par	Insp Par	AQL (See 4.3.2.1)
Electrical			
Receiver-Transmitter load protection	3.13.1.1	4.9.3.1	6.5% for the group
Random triggering rate	3.12.4	4.9.11	
Interrogation rates	3.12.11	4.9.17	
Receiver IF measurements:-	3.12.7	4.9.13	
IF frequency	3.12.7.1	4.9.13.1	
IF bandwidth	3.12.7.2	4.9.13.2	
IF pick-up	3.12.7.3	4.9.13.3	
Image response	3.12.8	4.9.14	
R-F Rejection	3.12.9	4.9.15	
Interchangeability	3.3	4.12	6.5%

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4.4.3 Group C inspection.- This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and B inspection.

4.4.3.1 Sampling for inspection of equipment.- One equipment per each 100 on order or 1 equipment per month, whichever of greater, shall be subjected to all tests and inspections per table IV, except that the one equipment shall be selected from the first production lot presented for acceptance inspection and sampling for accelerated life test shall be as specified in 4.4.3.2.

4.4.3.2 Five (5) AN/APX-44( ) per each 100 on order, or fraction thereof, shall be subjected to the following 250 hour accelerated life test to check compliance with 3.14:

a. Each AN/APX-44( ) to be tested shall be placed in full operation and allowed to operate continuously until failure occurs. (A failure is defined as any condition which precludes the performance of the intended mission of the transponder set.) Interrogation at a repetition rate of 1000, SIF, Reply, Mode 2, all pulses in (14 pulses).

4.4.3.3 Noncomplisnce.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

Table IV - Group C inspection

Inspection	Rqt Par	Test Par	Sampling (See 4.4.3.1)
Input power	3.6	4.8	
Undervoltage protection	3.6.1	4.8.1	
Overvoltage protection	3.6.2	4.8.2	
Transmitter frequency drift	3.11.5	4.9.2	
Receiver-Transmitter spurius response	3.13.8.3	4.9.5	
Receiver frequency drift	3.11.4	4.9.8	
Receiver frequency	3.11.2	4.9.7	
Transmitter frequency	3.11.2	4.9.1	
Triggering level drift	3.12.3	4.9.10	
Bandwidth drift	3.12.6	4.9.12.3	
Anti-jamming	3.12.10	4.9.16	
Other spurius responses	3.12.12	4.9.18	
Receiver video output	3.12.20	4.9.20.2	
Radar interference	3.12.17	4.9.22	
Echo effects	3.12.14	4.9.20	
Decoder Echo effects	3.12.15	4.9.20.1	
Power dissipation of oscillator crystal	3.8.4.1	4.9.25	
Frequency of crystal unit	3.8.4.2	4.9.26	
Antenna measurements	-	4.10	
Antenna mismatch	3.10.3	(4.10.1) (4.10.2)	
Antenna pattern	3.10	4.10.3	
Accelerated life test	3.14	4.14	(See 4.4.3.2)
Spike Suppression	3.12.19	4.9.23	
Suppression Characteristics	3.12.16.2	4.9.21	
Suppression Operation	3.12.16.3	4.9.21.1	
Transponder suppressed time	3.12.18	4.9.24	
Interference	3.4.6	4.16	

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4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.4.5 Disposition of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

(a) Sample units found defective during group A, B, or C inspection.

(b) Inspection lots which failed during group A or B inspection.

(c) Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Bounce preconditioning.- The receiver-transmitter and control unit with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The receiver-transmitter and control unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.6 Visual and mechanical inspection.-

4.6.1 Complete Transponder Set AN/APX-44( ).- The unit of the complete assembled Transponder Set AN/APX-44( ) shall be examined for defects listed in table V in addition to those listed in MIL-STD-252.

4.6.2 Classification of Defects.- Classification of defects major or minor, shall be determined by Table V below:

Table V - Classification of Defects
Printed Wiring Boards

Defect
MAJOR:
Bondage of Conductor to board: Any looseness of bond on any unsupported conductor length over 3/4 inch. Looseness of bond is greater than 1/8 inch on unsupported lengths under 3/4 inch. (Unsupported length is defined as that portion of the conductor between terminal points.)
Broken eyelet: Part of eyelet is missing. Multiple splits, two or more.
Seating of eyelet: Eyelet not properly seated perpendicular to the board.
Unplated Feed-thru hole: Plating is missing or incomplete in feed-thru holes.
Location of holes: Holes not entirely within area of terminal. (Edge of hole beyond edge of area of terminal (land) at any point.)

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Table V - Classification of Defects Printed Wiring Boards (Cont'd)

Defect
MAJOR: (Cont'd)
Superfluous conductor: Could cause potential short. Evaluate on basis of electrical clearance specified in Table VII.
Cuts, tears, or cracks in conductor: Deep or extensive enough to affect conductivity. Copper visible through plating and board surface visible through copper.
Reduction in area of conductor: Any form of reduction in area of conductor at any point allowing 50% or over reduction in width of conductor.
Terminal Area (land): View of board before soldering indicates that terminal area is completely missing (drilled away or never present) on any side of board containing conductor.
Peeling of conductor: Any indication of peeling of conductor. (Defect most prevalent at terminals and at ends of conductor contacts).
Conductor and lead spacing: Evaluate for potential short based on electrical clearances specified in Table VII.
Plating on conductors and conductor contacts: Lack of plating, when specified.
Cracks or chips in board: Any crack or chip which weakens conductor on board. Holes shall be free from cracks or chips. Cracks or chips on edges of board shall not be greater than 1/8 inch, except none allowed at contact ends.
Unspecified removal of board material: Any visible unspecified removal of board material (such as removal of material about conductors to increase insulation resistance - indication of contamination).
Warp or twist of board: Warp or twist in any board exceeds the specified allowed maximum.
Spacing of holes: Lack of proper spacing causes or could cause break through between holes or from holes to edges of board.
Protective insulating coating: Has not cured to full hardness (soft or tacky). Thickness is less than specified tolerance. This defect only applicable when such coatings are specified. The coating is applied after completion of mounting and soldering of all parts (Conductor contacts shall be free from such coating).
MINOR:
Bonding of conductor to board: Looseness of bond is 1/8 inch or less on unsupported lengths of conductor under 3/4 inch, including looseness at terminals.
Broken eyelet: Eyelet has no more than one split.
Seating of eyelet: Eyelet is loose (able to turn by hand).

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Table V - Classification of Defects Printed Wiring Boards (Cont'd)

Defect
MINOR: (Cont'd)
Location of holes: Distance between any edge of terminal area and edge of hole closer than minimum clearance dimension specified, but hole is entirely within area of terminal (land).
Cuts, cracks, or scratches in conductor: Board not visible through copper, but copper may be visible through plating when specified. Area of defect is completely across conductor or more than 1/2 inch along conductor (When this type of defect can be considered a reduction in area of conductor, evaluate in accordance with defects as applicable).
Reduction in area of conductor: Any form of reduction in area of conductors causing a reduction in width less than 50%, but exceeds specified tolerance (see Table VI).
Terminal area (land): Terminal area does not exceed by the specified dimension, the corresponding cross-sectional dimension of the hole.
Plating on conductors or contacts: Any part of contacts or conductors, when plating is specified, shows absence of plating or plating is less than minimum specified thickness.
Cracks or chips in board: Cracks or chips which may tend to weaken board, other than at conductors, holes, or edges of board (cracks or chips up to 1/8 inch allowed at edges of board, except at conductor contact edges).
Spacing of holes: Spacing between holes less than thickness of board. Spacing to edges of board less than twice the thickness of board. When spacings show evidence of breakthrough, evaluate in accordance with applicable defect. (Unless dimension for spacing is otherwise specified, in which case minimum tolerances apply.)
Protective insulating coating: Evidence of pinholes, bubbles, entrapped contaminates, etc., (when such coating is specified).
Size of holes: Size of holes exceed specified maximum cross-sectional dimensions.
Conductor pattern: Not clearly defined (ragged in spots on edges of conductor).
Loose force fit stud: Any force fit stud that can be turned or removed by hand. (This condition cannot be determined after soldering as the application of solder will cause the stud to be tight. This condition is unacceptable as it may cause a high resistance joint to develop which is very troublesome and difficult to locate after soldering.)

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Table VI - Conductor dimensions (based on 20°C temperature rise)

CURRENT		MINIMUM CONDUCTOR WIDTH
0.0014-inch-thick conductor	0.0028-inch-thick conductor	
Amperes	Amperes	Inch
1.5	2.5	0.020
2.5	3.5	0.030
3.5	4.7	0.060
4.7	8.3	0.125

Table VII - Conductor and lead spacing

Voltage (dc or peak ac)	CONDUCTOR SPACING		LEAD SPACING			
	COATED		UNCOATED, MIN			Lead to Printed Wiring
	Preferred	MIN		Clearance	Creepage	
Volts	Inch	Inch	Inch	Inch	Inch	Inch
Below 100	0.030	0.020	0.080	0.020	0.050	0.035
100 to 300	0.060	0.030	0.125	0.040	0.060	0.045
301 to 500	0.125	0.060	0.300	0.140	0.140	0.100

4.7 Performance.- The equipment shall be tested for all performance requirements of this specification and shall meet all requirements when operating under stated test conditions.

4.7.1 Test set-ups.- Transponder Set AN/APX-44( ) shall be assembled and cabled and power applied as required for performance of the following tests. The equipment shall be given a five (5) minute minimum warm-up time prior to performance of electrical tests.

4.7.2 Test conditions.- Unless otherwise specified, the test conditions shall be as specified in Specification MIL-E-5400.

4.8 Power input.- The Transponder Set AN/APX-44( ) shall be placed in full operation and the input current and voltage measured to determine compliance with 3.6.

4.8.1 Low voltage.- The equipment shall be subjected to an input voltage of 20 volts dc for a period of 30 minutes. The equipment shall resume normal operation when the input voltage is returned within the operating limits in accordance with 3.6.b..

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4.8.2 High voltage.- The equipment shall be subjected to an input voltage of 30 volts dc for a period of 30 minutes. The equipment shall resume normal operation when the input voltage is returned within the operating limits in accordance with 3.6b.

4.9 Receiver-transmitter.-

4.9.1 Transmitter frequency.- The transmitter frequency shall be measured at 5 mc/s intervals over the entire tuning range and shall comply with 3.11.2.

4.9.2 Frequency drift.- The transmitter frequency drift shall be measured and shall meet the requirements of 3.11.5b and 3.11.5c under the service conditions of 3.11.5a.

4.9.3 Power output.- Power output measurements shall be made on all modes of operation for compliance with 3.13.1 using a Polytechnic Universal power bridge type 650B(or equal) and a Polytechnic Research Bolometer Mount 628A (or equal).

4.9.3.1 Transmitter load protection.- The receiver-transmitter shall be turned on for 5 minutes for each condition specified in 3.13.1.1. After each one of the conditions have been met and the antenna or transmission line is properly connected to the receiver-transmitter, the receiver-transmitter shall give full specified performance in compliance with 3.13.1.1.

4.9.4 Transponder delay.- Properly interrogate the transponder with Radar Test Set AN/UPM-6B or equal at a frequency of 1030 megacycles per second. The interval of time between the 50% amplitude points on the leading edges of the second interrogation pulse and the first reply pulse shall be measured, using a Tektronic Oscilloscope type 545 (or equal) and a Tektronic Model 180 Time Mark Generator (or equal), and shall comply with 3.13.8.1.

4.9.5 Spurious response.- The equipment shall be tested for transmitter spurious response in compliance with 3.13.8.3, using Polarad Multipulse Spectrum Selector Model SD-1(or equal) and Polarad Spectrum Analyzer Model TSA-W (or equal).

4.9.6 Characteristics of reply.-

4.9.6.1 Pulse dimensions and characteristics.- Reply pulse dimensions and characteristics shall be viewed on a Tektronic Oscilloscope type 545 (or equal) utilizing the following procedure, to determine compliance with the following requirements:

a. Pulse shape.- The transponder set shall be interrogated at 1030 megacycles per second and the demodulated transmitter pulses shall be viewed to determine compliance with 3.13.6 and 3.13.2.

b. Pulse spacing.- Properly interrogate the transponder set at 1030 megacycles per second and view the demodulated transmitter pulses for compliance with 3.13.2, 3.13.2.1 to 3.13. 3.2, 3.13.4 to 3.13.6.2 and 3.13.7. A Tektronic Model 180 Time Mark Generator (or equal) shall be used as a reference standard.

4.9.6.2 Code number assignment.- Properly interrogate the transponder set at 1030 megacycles per second and view the demodulated transmitter pulses on a Tektronic oscilloscope type 545 (or equal) to determine that the position of information pulses on each mode and assigned code number as shown on code switches comply with 3.13.8.4 through 3.13.8.4c for following code and mode settings: Mode 1 codes 00, 01, 02, 03, 13, 23, 33, 43, 53, 63 and 73. Mode 2 codes 0000, 1111, 2222, 3333, 4444, 5555, 6666, 7777. Mode 3 codes 00, 01, 02, 03, 04, 05, 06, 07, 17, 27, 37, 47, 57, 67, 77.

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4.9.6.3 Special reply pulse.- With a Beckman/Berkley Double Pulse Generator Type 903 or 904 (or equal) connected to a Hewlett-Packard UHF Signal Generator Type 612A or 614A (or equal), and with the output of the signal generator and the demodulated R.F. pulses viewed on a Tektronix Oscilloscope Type 545 (or equal), at the trigger-in receptacle, apply to the transponder pulse-paired trigger pulses spaced 4.0 microseconds apart and at 1000 pulse-pairs per second to determine compliance with 3.13.6.2.1.

4.9.6.4 Reply operation.- Reply pulses shall be viewed on a Tektronix Oscilloscope Type 545 (or equal) utilizing the following procedure, to determine compliance with the following requirements:

4.9.6.4.1 Mode 1 reply.- With the controls specified in 3.9.2b set for Mode 1 operation, properly interrogate the receiver-transmitter within the specified frequency range of the equipment and view the demodulated transmitter pulses in accordance with the following:

a. Set the control specified in 3.9.2e to the "NORMAL" position and view the reply to determine compliance with 3.13.2.1.

b. Set the control specified in 3.9.2e to the "MOD" position and view the reply to determine compliance with 3.13.4.1.

4.9.6.4.2 Mode 2 reply.- With the control specified in 3.9.2b set for Mode 2 operation, properly interrogate the receiver-transmitter within the specified frequency range of the equipment and view the demodulated transmitter pulses in accordance with the following:

a. Set the control specified in 3.9.2e to the "NORMAL" position and view the reply to determine compliance with 3.13.2.2.

b. Set the control specified in 3.9.2e to the "MOD" position and view the reply to determine compliance with 3.13.4.3.

4.9.6.4.3 Mode 3 reply.- With the controls specified in 3.9.2b set for Mode 3 operation, properly interrogate the receiver-transmitter within the specified frequency range of the equipment and view the demodulated transmitter pulses in accordance with the following:

a. Set the control specified in 3.9.2e to the "NORMAL" position and view the reply to determine compliance with 3.13.2.3.

b. Set the control specified in 3.9.2e to the "MOD" position and view the reply to determine compliance with 3.13.4.3.

4.9.6.4.4 Emergency reply.- With the controls specified in 3.9.2a set for "EMER" operation, properly interrogate the receiver-transmitter within the specified frequency range of the equipment and view the demodulated transmitter pulses in accordance with the following:

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4.9.6.4.4 Emergency reply.- (Cont'd)

a. Set the control specified in 3.9.2e to the "NORMAL" position and view the reply to determine compliance with 3.13.2.4.

b. Set the control specified in 3.9.2e to the "MOD" position and view the reply to determine compliance with 3.13.5.

4.9.6.4.5 I/P reply.- Properly interrogate the receiver-transmitter within the specified frequency range of the equipment, then using the control specified in 3.9.2d and in accordance with the following:

a. Set the control specified in 3.9.2e to the "NORMAL" position and view the demodulated reply pulses to determine compliance with 3.13.3.

b. Set the control specified in 3.9.2e to the "MOD" position and view the demodulated reply pulses to determine compliance with 3.13.5.

4.9.6.5 Civil identification.- With the transponder properly interrogated and the controls selected as specified in 3.13.7, the reply shall be viewed on a Tektronix Oscilloscope Type 545 (or equal) and shall be as specified in 3.13.7.

4.9.7 Receiver frequency.- Receiver frequency shall be measured at 5 mc/s intervals over the entire tuning range and shall comply with 3.11.2.

4.9.8 Receiver frequency drift.- The receiver frequency drift shall be measured and shall comply with 3.11.4.

4.9.9 Triggering level.- With the transponder set properly interrogated (interrogation pulse separation adjusted to give optimum triggering level and interrogation pulse is within the limits specified in 3.11.1) at 1030 megacycles per second by a signal generator and a Tektronix Oscilloscope type 545 (or equal) connected to the suppression receptacle "SUPPR" on the front panel of the receiver-transmitter adjust the oscilloscope to the point where reply trains appear. Reduce signal generator output until two different traces appear on the oscilloscope. A pulsed trace appears when firing occurs and a straight trace appears when firing does not occur. When the two different traces occur this is partial firing. Fifty (50) percent firing is defined as that partial firing where the intensity of the straight trace is equal to the intensity of the pulsed trace. The triggering level shall be measured and shall comply with 3.12.1. (See 6.4.15).

4.9.9.1 Triggering level selection.- The transponder set shall be operated utilizing the following procedure to determine compliance with 3.12.2:

a. Preset the low triggering level to 42 dbm.

b. Set the control specified in 3.9.2a to the "LOW" position.

c. Properly interrogate transponder set within the operating frequency range of the equipment. Check for compliance with 3.12.2.

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4.9.10 Triggering level drift.- With the triggering level set to a nominal value, the triggering level drift shall be measured and shall comply with 3.12.3.

4.9.11 Random triggering rate.- With the 51-ohm resistive load connected to the antenna terminal, with the absence of external signals and with the equipment set for normal triggering level, the random triggering rate shall be measured and shall be in accordance with 3.12.4. These measurements shall be conducted with the equipment in an R.F. shielded enclosure.

4.9.12 Bandwidth measurements.-

4.9.12.1 Overall receiver triggering bandwidth.- The transponder set shall be tested in accordance with the following to determine compliance with 3.12.5:

4.9.12.2 Triggering bandwidth.- Properly interrogate the equipment at 1030 megacycles per second and reduce the interrogating signal level to the 50% firing point. Increase the interrogating signal by 3 db and measure the frequency to both sides of the center frequency at which the 50% firing point is again reached. The difference in frequency between the two 50% firing points shall be measured and shall be as specified in 3.12.5. The frequencies at which the response is 40 db down from maximum response shall be measured and shall comply with 3.12.5.

4.9.12.3 Bandwidth drift.- The bandwidth drift of the receiver shall be measured and shall meet the requirements of 3.12.6.

4.9.13 I. F. measurements.-

4.9.13.1 I.F. frequency.- The center frequency of the IF amplifier shall be measured and shall comply with 3.12.7.1.

4.9.13.2 I.F. bandwidth.- With the output of a sweep type signal generator fed into the input of the IF amplifier, and the output of the IF amplifier fed into an oscilloscope, adjust the signal generator to produce an RF output which sweeps approximately 6 mc/s around the IF frequency. The difference between the two frequencies at which the response is down 3 db from maximum response shall be as specified in 3.12.7.2.

4.9.13.3 I.F. pick-up.- The IF pick-up shall be measured and shall comply with 3.12.7.3.

4.9.14 Image response.- The image response to signals of image frequencies shall be measured and shall comply with 3.12.8.

4.9.15 R.F. rejection.- With the receiver tuned to 1010, 1020 and 1030 megacycles per second the R.F. rejection shall be measured and shall comply with 3.12.9.

4.9.16 Anti-jamming.- With interfering signals of simple CW, AMCW, random-spaced pulses, and regularly-spaced pulses the equipment shall be tested as follows to determine compliance with 3.12.10:

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4.9.16 Anti-jamming.- (Cont'd)

a. Properly interrogate the equipment at triggering levels 3 db and 6 db above the normal triggering level. Apply interfering signals (as described above) at triggering levels 7 and 4 db down from the normal triggering level. The reply pulses shall be viewed on a Tektronic Oscilloscope type 545 (or equal) and shall indicate at least a 50% firing level.

b. Properly interrogate the equipment at triggering levels 20, 23, 28 and 30 db above the normal triggering level. Apply interfering signals (as described above) at levels 17, 20, 25 and 27 above the normal triggering level. The reply pulses shall be viewed on a Tektronic Oscilloscope type 545 (or equal) and shall indicate at least a 50% firing level.

4.9.17 Interrogation rates.- Properly interrogate the equipment with 15 to 500 interrogations per second. The triggering signal level for each interrogation rate shall be measured. With the sensitivity-reduction control adjusted to 500, 1000 and 2000 interrogations per second, properly interrogate the equipment with interrogations 50 to 100% greater than the sensitivity-reduction control and measure the sensitivity below normal triggering level. The equipment shall perform in accordance with 3.12.11.

4.9.18 Other spurious responses.- The other spurious response shall be in compliance with 3.12.12.

4.9.19 Decoder.- With a Beckman/Berkley Double Pulse Generator Type 903 or 904 (or equal) connected to a Hewlett-Packard UHF Signal Generator Type 612A (or equal), and with the output of the signal generator viewed on a Tektronic Oscilloscope type 545 (or equal), apply calibrated 1030 megacycle per second interrogations to the equipment in accordance with 3.12.13 and the following:

4.9.19.1 Decoder tolerances - Mode 1.- With the equipment set for Mode 1 operation, the upper and lower decoder 50% firing point limits shall be measured and shall comply with 3.12.13.1.

4.9.19.2 Decoder tolerances - Mode 2.- With the equipment set for Mode 2 operation, the upper and lower decoder 50% firing point limits shall be measured and shall comply with 3.12.13.2.

4.9.19.3 Decoder tolerances - Mode 3.- With the equipment set for Mode 3 operation, the upper and lower decoder 50% firing point limits shall be measured and shall comply with 3.12.13.3.

4.9.19.4 Decoder tolerances - Modes 2 and 3.- With the equipment set for simultaneous operation of Modes 2 and 3, the upper and lower decoder 50% firing point limits shall be measured and shall comply with 3.12.13.4.

4.9.19.5 Decoder - single pulse.- Interrogate the equipment with single pulse interrogation signals of various pulse durations and the equipment shall not respond in compliance with 3.12.13.5.

4.9.19.6 Decoder control.- The operation of remote controls of the equipment shall comply with 3.12.13.6.

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4.9.20 Echo effects.- The equipment shall operate in the presence of echoes of the interrogating signals as specified in 3.12.14. To test the equipment, the echo may be simulated by a signal of slightly different frequency (but not greater than one megacycle difference) than the interrogation.

4.9.20.1 Echo effects - decoder.- The transponder shall meet the requirements of 3.12.15 when operating in the presence of echoes as specified in 3.12.14.

4.9.20.2 Receiver video output.- The receiver video output shall be in accordance with 3.12.20.

4.9.21 Suppression.- Apply a suppression pulse having the characteristics specified in 3.12.16.2 to the transponder. The equipment shall operate as specified in 3.12.16.3 when tested as specified in 4.9.21.1.

4.9.21.1 Operational test.- The equipment shall be such that it is rendered insensitive to a 1030 megacycle RF pulse input of 10 volts on the antenna terminal for the duration of the suppression pulse and for a period of at least 15 microseconds after the termination of the suppression pulse.

4.9.21.2 Suppression receptacle.- The output from the suppression receptacle shall be viewed on an oscilloscope and the characteristics viewed shall indicate a proper suppression signal in compliance with 3.12.16.1.

4.9.22 Radar interference.- Radar interference with the equipment shall be measured and shall be in accordance with 3.12.17.

4.9.23 Spike suppression.- Apply spike pulses less than 0.3 microseconds in duration. The equipment shall meet the requirements of 3.12.19.

4.9.24 Transponder suppressed time.- Properly interrogate the equipment and measure the transmitter reply to any other interrogation during the reply pulse train. The suppressed time shall be as specified in 3.12.18.

4.9.25 Power dissipation of oscillator crystal.- The power dissipation of the crystal unit shall be measured and shall comply with 3.8.4.1.

4.9.26 Frequency of crystal unit.- The frequency limits of the crystal unit shall be measured and shall comply with 3.8.4.2.

4.9.27 Range jitter.- Properly interrogate the equipment and measure the range jitter. The range jitter shall comply with 3.13.8.2.

4.10 Antenna measurements.-

4.10.1 Antenna mismatch.- With Antenna AT-484( )/APX-44 connected as shown in Figure 1, set the signal generator at 1090 megacycles per second, and adjust the slotted line and broad-band probe to obtain maximum reading on the standing wave amplifier. Adjust the signal generator and amplifier to indicate a zero reading on the decibel scale. With the slotted line readjusted for minimum reading on the standing wave amplifier, the db read directly on the meter shall meet the requirements of 3.10.3. The above test shall be repeated at 1095, 1100, 1105, 1110, 1020, 1025 and 1030 megacycles per second.

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4.10.2 Voltage standing wave ratio (VSWR).- The voltage standing wave ratio shall be measured and recorded whenever the transmitter frequency check is made during the service conditions specified in 3.11.5 and shall be no greater than 1 4 :1.

4.10.3 Antenna pattern.-

a. With the equipment connected as shown in Figure 3, excite the antenna with a 1000 mc/sec., 0 dbm, signal.

b. Position the leading edge of the antenna, so that it points directly at the pick-up dipole. This shall be the zero degree point for horizontal readings.

c. Adjust the field strength meter for relative carrier readings and record the reading for the antenna zero degree position. Rotate the antenna clockwise 30° and record the relative field strength reading. Repeat for all 30° positions.

d. Reposition the antenna at zero degrees and record the field strength.

e. Tilt the antenna 30° in the vertical plane and record the relative field strength. Repeat for the 60°, 90°, 150° and 180° positions.

f. Plot all readings above on polar graph paper. The graphs shall exhibit the characteristics of the antenna specified in 3.10.

4.11 Service conditions.-

4.11.1 Temperature and altitude, combination.- The equipment shall be subjected to the temperature-altitude combination tests as specified in Specification MIL-E-5422 to determine compliance with 3.4.7.2. Measurements shall be taken as follows:

Paragraph	Steps	Measurements	Test Results
a. 4.8	(Test set-up)	(At Input Voltages):	3.6
b. 4.9.2	1 through 14 inclusive	Transmitter Frequency Drift	3.11.5
c. 4.9.17		Interrogation rate	3.12.11
d. 4.10.2	(same as 4.11.1b)	Antenna (VSWR)	3.10.3
e. 4.9.8	1,3 through 10 inclusive	Receiver Frequency Drift	3.11.4
f. 4.9.12.3	1,6,7 and 9	Bandwidth drift	3.12.6
g. 4.9.10	1 through 14 inclusive	Triggering Level Drift	3.12.3

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4.11.2 Humidity.- The equipment shall be subjected to the humidity test as specified in Specification MIL-T-5422 to determine compliance with 3.4.7.3.

4.11.2.1 Performance.- Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurement	Performance during cycle (Step) (Test Set-up)
a. 3.6b	Input voltage (29V. DC)	
b. (3.11.2)	(Transmitter frequency)	
(3.11.5)	(Transmitter frequency drift)	4 and 6
c. 3.11.4	Receiver frequency drift	4 and 6
d. 3.12.3	Triggering level drift	4 and 6
e. 3.10.3	Antenna (VSWR)	4 and 6

4.11.3 Vibration.- The equipment shall be subjected to the vibration tests of Specification MIL-E-5422, to determine compliance with 3.4.7.4. Measurements during tests shall be as specified herein, with the equipment positioned during the vibration tests as specified in 4.11.3.1.

4.11.3.1 Normal installation orientation during vibration test.-

a. Receiver-Transmitter RT-494( )/APX-44 fully secured in the intended manner to Mount MT-2100( )/APX-44.

b. Control, Transponder C-2714( )/APX-44 fully secured in a manner with the lighted panel surface in a vertical plane.

c. Antenna AT-884( )/APX-44 fully secured in a manner with the adapter flange in a horizontal plane and the blade extending downward from that plane.

4.11.3.2 Receiver-Transmitter RT-494( )/APX-44 installed on Mount MT-2100( )/APX-44.-

Paragraph	Measurement	Test Procedure
a. 3.6b	Input Power (27.5V. DC)	(Set-up)
b. 3.12.1	Triggering Level)	
c. 3.12.5	Triggering bandwidth)	Step 2
d. 3.13.1	Output power	

4.11.3.3 Receiver-Transmitter RT-494( )/APX-44 without Mount MT-2100( )/APX-44.- Measurements shall be made during this vibration test, consisting of types specified in 4.11.3.2a through 4.11.3.2d inclusive.

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4.11.3.4 Antenna AT-884( )/APX-44.- Measurements during vibration test shall be as follows:

Paragraph	Measurement	Test Procedure
3.10.3	Antenna (VSWR)	Step 2

4.11.3.5 Control Transponder C-2714( )/APX-44.- Measurements during vibration test shall be as specified in paragraph 4.11.3a through 4.11.3d inclusive.

4.11.4 Shock.- With the receiver-transmitter installed on Mount MT-2100( )/APX-44, the equipment shall be subjected to the sand-drop test as specified in Specification MIL-E-5422. Measurements as listed below shall be made and recorded to determine compliance with requirements of 3.4.7.5.

Measurement	Paragraph
Normal triggering level	3.12.1
Triggering bandwidth	3.12.5
Power Output	3.13.1
Reply pulse width	3.13.2

4.11.4.1 Mount MT-2100( )/APX-44.- With a dummy load substituted in mounting, the mount shall be subjected to the tests of Mountings (crash-safety) of Specification MIL-E-5422. After completion of test the mounting base and vibration isolators shall be examined for mechanical failure.

4.11.5 Explosion-proof test.- The transponder set shall be subjected to the explosion-proof test of Specification MIL-E-5422, except, that test procedure for operation of all making and breaking electrical contacts shall not apply to the electrical switches on the front panel of Receiver-Transmitter RT-494( )/APX-44, during operation of the equipment.

4.11.6 Rain.- Antenna AT-884( )/APX-44 shall be subjected to the rain test of Specification MIL-T-5272 to determine compliance with 3.4.7.7. Measurements shall be taken as follows, before and after rain tests:

Paragraph	Measurement	Test Results
3.10.3	Antenna (VSWR)	4.10.2

4.11.7 Sand and dust test.- The transponder set shall be subjected to the sand and dust tests of Specification MIL-E-5422. Measurements shall be taken at conclusion of tests consisting of the following:

Paragraph	Measurement
3.12.1	Normal triggering level
3.12.5	Triggering bandwidth
3.13.1	Power Output
3.13.2	Reply Pulse Characteristic
3.11.2	Transmitter frequency

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4.12 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.3. When a listed dimension is not within specified or design limits it shall be considered a major defect.

4.12.1 Mount MT-2100( )/APX-44.-

a. Check location of guide (hold-down) pins, hold-down clamps, and index pins.

b. Check location of connector cut-out for connector plug.

c. Check minimum width and depth of mount.

d. Check location of injector-ejector mechanism which receives mating receiver-transmitter.

e. Check location of mounting holes on shock mount.

4.12.2 Receiver-Transmitter RT-494( )/APX-44.-

a. Check location of hold-down hook and the handle which receives the hold-down clamps of the mount.

b. Check maximum outside dimensions of length and width of case.

c. Check location of the cable connector receptacle.

4.13 Tests for protective coating applied to printed wiring circuit boards.- The characteristics of the protective coating to be used shall be determined by the sample boards prepared from the copper-clad laminate material to be used in the construction of the equipment. The sample board shall be 1-1/2 inches wide by 4-1/2 inches long, and shall bear a wiring pattern centered on one side consisting of six sharply defined parallel lines .030 inch ±.003 inch wide, spaced .030 inch ±.003 inch apart and 3 inches in length. Alternate lines shall be connected to terminal areas at opposite ends of the pattern (see Figure 1); also attached to the terminal area shall be wire leads to be used in testing. The sample boards shall be prepared by the use of the same processes (etching, fluxing, soldering, cleaning, etc) to be used in the production of the equipment. The coating material shall be applied to the sample board and cured. Prior to testing, all coating material shall be removed from the lengthwise edges of the board if the edges were not masked prior to coating.

4.13.1 Thickness of protective coating.- The thickness of the coating shall be measured on four unencumbered flat areas of the board using Method 618.1 of Specification TT-P-141. The micrometer used shall be accurate to .001 inch.

4.13.2 Curing time and temperature.- Curing time and temperature tests will be performed under the applicable conditions of section 200 of Specification TT-P-141.

4.13.3 Temperature cycling.- Temperature cycling tests shall be conducted in accordance with condition D, Method 102A of Standard MIL-STD-202. Five cycles shall be conducted.

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4.13.4 Moisture resistance.- Moisture resistance tests shall be conducted in accordance with Method 106 of Standard MIL-STD-202. Ten moisture resistance test cycles shall be conducted, with a polarizing voltage of 100 volts applied to each specimen . The low temperature and vibration steps shall be omitted. Insulation resistance measurements shall be made before cycling and after the first, fourth, seventh, and final cycles in the manner described in 4.13.5.

4.13.5 Insulation resistance.-

4.13.5.1 Before moisture resistance.- Insulation resistance tests shall be conducted in accordance with Method 302 of Standard MIL-STD-202, using test condition A (100 volts test potential). All measurements shall be taken after an electrification period of 1 minute.

4.13.5.2 During moisture resistance.- Upon completion of step 6 of the first, fourth, seventh, and final cycles of the moisture resistance test, the specimens shall be conditioned at a temperature of 77° ±3°F and a relative humidity of 90-95 percent for a period of 1-1/2 to 3-1/2 hours. Insulation resistance measurements shall then be made under these conditions using the procedure of 4.13.5.1.

4.13.5.3 After moisture resistance.- Following step 6 of the final cycle, the test specimens shall be maintained at a temperature of 77° ±3°F and a relative humidity of 45-55 percent for a period of 24 hours, after which insulation resistance measurements shall be made using the procedure of 4.13.5.1.

4.13.6 Dielectric strength.-

4.13.6.1 Before temperature cycling and moisture resistance.- The dielectric strength test shall be conducted in accordance with Method 301 of Standard MIL-STD-202. All measurements shall be made using a test voltage of 1500 volts rms at 60 cps.

4.13.6.2 After temperature and moisture resistance cycling.- Upon completion of temperature cycling or step 6 of the final cycling of the moisture-resistance test, the test specimens shall be maintained at a temperature of 77° ±3°F and relative humidity of 45-55 percent for a period of 24 hours, after which dielectric strength tests shall be conducted using the procedure of 4.13.6.1.

4.14 Accelerated life test.-

4.14.1 Test.- Each AN/APX-44( ) to be tested (see 4.4.3.2) shall be placed in full operation and allowed to operate continuously until a failure occurs. (A failure is defined as any condition which precludes the performance of the intended mission of the Transponder Set). Operation of the equipment shall be as follows: Constant interrogation at a repetition rate of 1000, SIF reply, Mode 2, all pulses in. (14 pulses).

4.14.2 Failure.- In the event of a failure, the contractor shall immediately determine its cause and make the necessary repairs.

4.14.3 Total hours of operation.- Immediately after the repairs are made the transponder sets shall be returned to operation until each equipment has been in operation for a total of 250 hours.

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4.15 Procurement inspection of preparation for delivery.- Inspection of preparation for delivery shall conform to Specification MIL-P-116.

4.16 Interference.- The transponder set shall be tested in accordance with the tests of Specification MIL-I-6181 to determine compliance with 3.4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing and packaging and marking shall conform to Drawing SC-A-362100.

6. NOTES

6.1 Use.- The equipment is used in Army aircraft to carry our specific military missions.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

c. Preproduction inspection.-

(1) Two samples of each item cited in section 1 are generally required. (See 3.17).

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required. (See 3.16)

g. Quantity of tools and running spare parts required. (See 3.16).

h. Facsimiles required. (See 3.4.4.1)

i. Submission of the statement of treatment referenced in 3.4.3.2, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/APX-44W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.4 Definitions.- The following definitions shall apply to this specification:

6.4.1 Rise time.- Rise time is the time required for the voltage to increase from 10% to 90% of peak amplitude on the leading edge of the pulse.

6.4.2 Fall or decay time.- The time required for the voltage to decrease from 90% to 10% of peak amplitude on the trailing edge of the pulse.

6.4.3 Pulse width or duration.- Pulse width is the time interval between the 50% amplitude points on the leading and trailing edges of the pulse.

6.4.4 Delay and spacing.- Delay and spacing between pulses is the time interval between the 50% amplitude points on the leading edges of the pulses involved.

6.4.5 Coder.- A circuit which generates or forms a coded reply in a transponder or an interrogation code in an interrogator-responder.

6.4.6 Decoder.- A circuit which responds to a particular interrogation mode and rejects all others.

6.4.7 Echo suppression.-A form of transponder circuit operation requiring successive pulses to be nearly the same amplitude to cause triggering.

6.4.8 Interrogation.- An RF pulse or a combination of pulses intended to trigger a transponder.

6.4.9 Jitter.- A momentary variation in pulse separation, expressed in microseconds.

6.4.10 Pulse amplitude.- The maximum voltage amplitude of the pulse or the rf pulse envelope, excluding spike.

6.4.11 Pulse amplitude jitter.- The variation of the instantaneous pulse amplitude of successive pulses caused by unintentional effects within the equipment, expressed in terms of percent of average instantaneous amplitude.

6.4.12 Pulse separation.- The interval of time, expressed in microseconds between 50 percent pulse amplitude points on the leading edges of the pulses involved.

6.4.13 Reply.- An rf pulse group transmitted by a transponder as a result of an interrogation.

6.4.14 Transponder delay.- The interval of time, expressed in microseconds, between the last component RF pulse of the interrogation and the first component RF pulse of the reply, measured between the 50 percent pulse amplitude points on the edge of the pulses involved.

6.4.15 Receiver triggering level.- The amplitude of an interrogation pulse required to just cause 50 percent firing of the equipment, with the interrogation pulses being supplied by a signal generator having a nominal 51 ohm output impedance.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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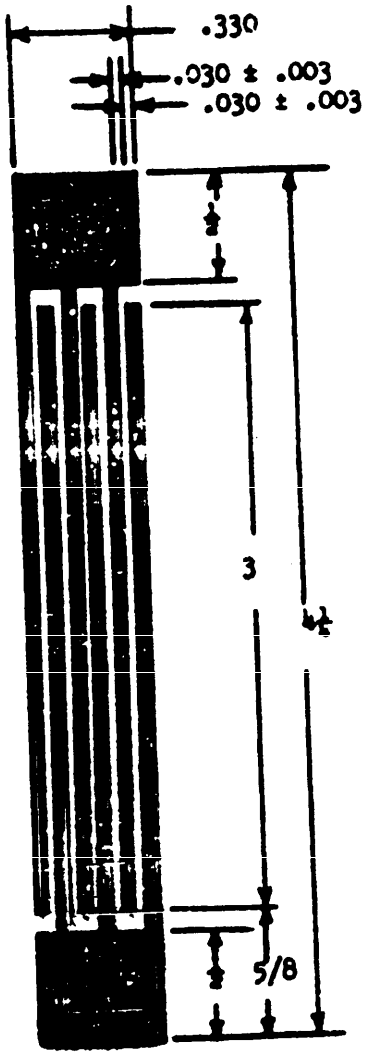


Figure 1

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FIGURE 2 ANTENNA (VSWR) TEST SET-UP

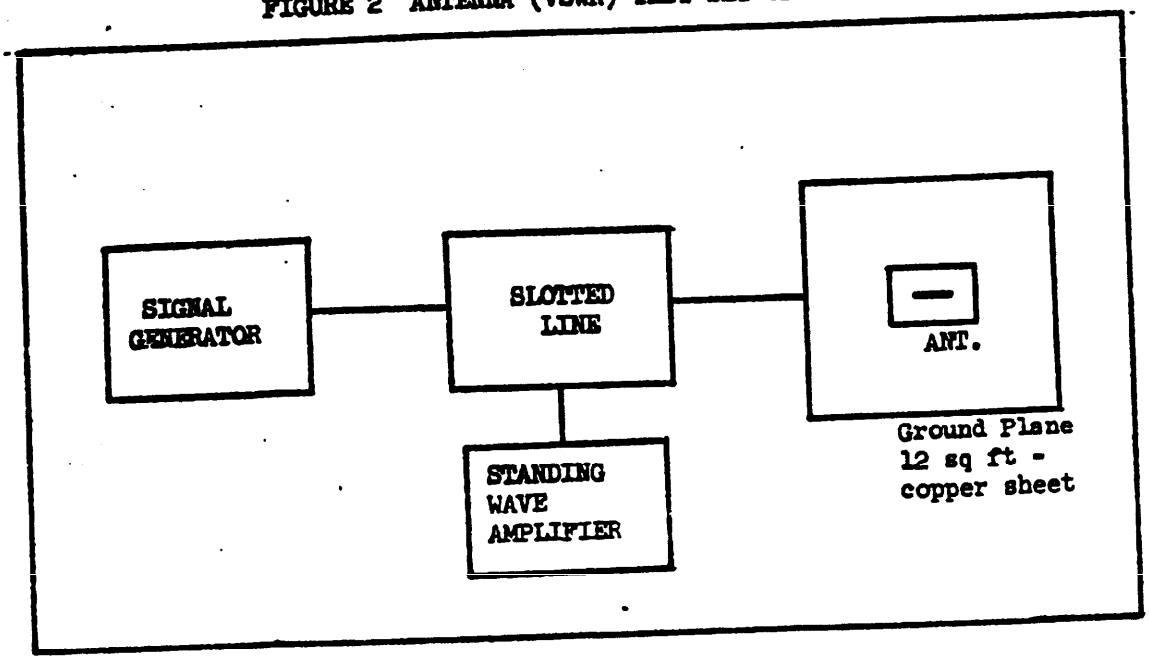
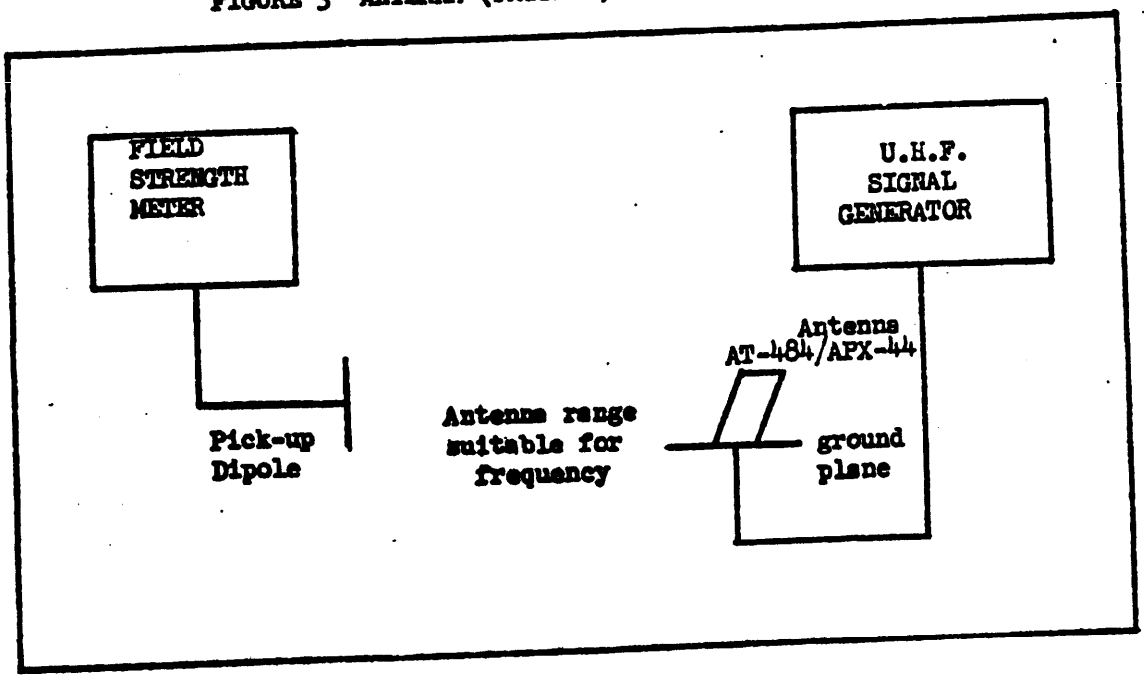


FIGURE 3 ANTENNA (PATTERN) TEST SET-UP



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